

Cybersecurity Vulnerability Assessment Report

Submitted By

Name: Meghana Nagapuri

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Domain: Cybersecurity

Task: Vulnerability Assessment Using Nmap and Nikto

1. Objective

The objective of this task is to perform a basic vulnerability assessment on a target server using cybersecurity tools to identify open ports, services, and vulnerabilities.

2. Tools Used

- Nmap – Network scanning and service detection
- Nikto – Web server vulnerability scanning
- Windows 11 Laptop – Execution environment

3. Target Used

scanme.nmap.org

This is a legal testing server provided by Nmap for educational purposes.

4. Nmap Scan

Command Used:

```
nmap -sV scanme.nmap.org
```

Purpose:

- Detect open ports
- Detect running services
- Identify service versions

Nmap Findings

- Port 22 – SSH
- Port 80 – HTTP

Security Concerns:

- Outdated Apache version

- Older OpenSSH version
- Exposed services increase attack surface

5. Nikto Scan

Command Used:
`nikto -h scanme.nmap.org`

Findings:

- Missing security headers
- Outdated Apache server
- HTTP methods enabled
- Information disclosure

6. Risk Analysis

Outdated Apache Version:
Older Apache versions may contain known vulnerabilities.

Outdated OpenSSH Version:
Older SSH versions may contain security weaknesses.

Missing Security Headers:
Missing HTTP headers reduce protection against browser-based attacks.

7. Mitigation Steps

- Update Apache regularly
- Use strong passwords and authentication
- Configure secure HTTP headers
- Disable unnecessary services and ports

8. How to Perform This Task

Step 1: Install Nmap
Download from: <https://nmap.org/download.html>

Step 2: Install Kali Linux or Nikto
Download from: <https://www.kali.org/get-kali/>

Step 3: Run Nmap Scan
Open Command Prompt and run:
`nmap -sV scanme.nmap.org`

Step 4: Run Nikto Scan
Open Kali Linux terminal and run:

```
nikto -h scanme.nmap.org
```

Step 5: Save Results

Copy the outputs into text files and prepare the report.

9. Final Result

The vulnerability assessment task was successfully completed using Nmap and Nikto tools. Open ports, running services, outdated software versions, and web server vulnerabilities were identified and analyzed. Recommendations were provided to improve system security.

10. Conclusion

This task demonstrated how cybersecurity tools are used to identify vulnerabilities and security weaknesses in systems. Regular scanning and proper mitigation methods help improve cybersecurity protection.

11. Viva Questions and Answers

Q1. What is Nmap?

Nmap is a network scanning tool used to identify hosts, ports, and services.

Q2. What is Nikto?

Nikto is a web server vulnerability scanner.

Q3. Why are open ports important?

Open ports may become entry points for attackers.

Q4. What is vulnerability assessment?

It is the process of identifying security weaknesses in a system.

12. References

- Nmap Official Documentation
- Nikto Documentation
- OWASP Security Guidelines